

Chapter 5: Results

5.1 Experimental overview

We trained NEAT and HA-NEAT on the same multi-task problem: a single evolved network must control both a Hopper (11 observations, 3 actions) and a Walker2D (17 observations, 6 actions). The network is sized to the larger task. Hopper observations are zero-padded to 17 dimensions, and the 6-dimensional output is sliced to 3 for Hopper’s actuators. Fitness is the normalized minimum across both tasks, i.e. the worse of the two normalized per-task rewards. A network that learns Hopper well but neglects Walker2D scores poorly.

Three experiments make up the comparison. First, the main head-to-head: NEAT vs HA-NEAT across population sizes, generation counts, and random seeds. Second, a multi-episode re-evaluation of saved genomes, because training fitness comes from a single episode and may not reflect actual policy quality. Third, an ablation that runs HA-NEAT’s infrastructure (OriginConn, HANEAT-Mutation) with a single activation function, testing whether any advantage comes from activation diversity or from the machinery around it. Section 5.2 reports aggregate performance, Section 5.3 breaks it down per task, Section 5.4 compares training estimates to re-evaluation, Section 5.5 covers species dynamics and genome complexity, Section 5.6 presents the ablation, and Section 5.7 justifies why we used normalized-minimum aggregation.

All runs used the MJX (MuJoCo on GPU) physics backend, population sizes of 1024–4096, and between 500 and 1000 generations. [PLACEHOLDER: final hardware details — GPU type, JIT compile time, wall-clock per generation]. Random seeds were drawn from {42, 123, 75, 7, 21}; preliminary results use subsets of 2–3 seeds, and the final experiment uses all 5. Reference rewards used for normalization are 3000 (Hopper) and 5000 (Walker2D), drawn from BRAX_REFERENCE_REWARDS.

5.2 Multi-task performance comparison

We compared the best-found normalized-minimum fitness per run across both methods and several configurations. Table 1 collects the results across experiment versions; Figure 1 shows the fitness training curves for the final experiment.

Table 1. Aggregate normalized-minimum fitness per method across experiment versions. Best-genome fitness reported as saved in the .npz checkpoint (single-episode training estimate). – indicates missing runs.

Experiment Method		Pop	Gens	Seeds	Scores (per seed)	Mean
v3	NEAT	2048	500	42, 123	0.358, 0.341	0.349
v3	HA-NEAT	2048	500	42, 123	0.342, 0.484	0.413
v4	NEAT	4096	1000	42	0.399	—
v4	HA-NEAT	4096	1000	42, 123	0.473, 0.457	0.465
final	NEAT	4096	2000	42, 123, 75, 7, 21	[PLACEHOLDER]	[PLACEHOLDER]
final	HA-NEAT	4096	2000	42, 123, 75, 7, 21	[PLACEHOLDER]	[PLACEHOLDER]

In the preliminary v3 runs (pop=2048, 500 generations), HA-NEAT reached a mean aggregate fitness of 0.413 across two seeds, compared to 0.349 for NEAT — a difference of 0.064 normalized units. This difference is driven primarily by one high-performing HA-NEAT run (seed 123: 0.484). In the v4 configuration (pop=4096, 1000 generations), the single completed NEAT run achieved 0.399, while the two HA-NEAT runs averaged 0.465. The final experiment, with 5 seeds per method, will determine whether this advantage holds across a broader sample. [PLACEHOLDER: final experiment mean $\pm$ std for NEAT and HA-NEAT. Update this paragraph once complete.]

Across both experiment versions, HA-NEAT tended toward higher aggregate fitness, though with high cross-seed variance in both methods. The aggregate number alone does not reveal whether both tasks improved or whether one task carried the other.

5.3 Per-task analysis: Hopper and Walker2D

A moderate aggregate score can hide opposite extremes: a network that excels on Hopper but fails Walker2D, or vice versa, still reports a nonzero normalized minimum. We split the aggregate into per-task fitness to check whether the two methods differ in which task they learn and which they neglect.

Table 2. Per-task fitness for best genomes in v4 experiment. Training best = single-episode estimate from MLflow `best_fitness/<task>`. Re-eval = mean $\pm$ std over 5 episodes (see Section 5.4). Normalized = raw / reference reward (Hopper: 3000, Walker2D: 5000).

Method	Seed	Hopper (train)	Hopper (re-eval)	Walker2D (train)	Walker2D (re-eval)
HA-NEAT	42	1905 (0.635)	1860 $\pm$ 322 (0.620)	2365 (0.473)	2005 $\pm$ 162 (0.401)
HA-NEAT	123	1370 (0.457)	752 $\pm$ 435 (0.251)	2330 (0.466)	1953 $\pm$ 483 (0.391)
NEAT	42	1200 (0.400)	1196 $\pm$ 3 (0.399)	1997 (0.399)	1987 $\pm$ 11 (0.397)

NEAT seed42 produced remarkably balanced per-task scores: 1200 on Hopper (0.400 norm) and 1997 on Walker2D (0.399 norm), nearly identical normalized values. HA-NEAT seed42 looked different. It reached 1905 on Hopper (0.635 norm) but only 2365 on Walker2D (0.473 norm). Walker2D was the bottleneck in the normalized-minimum aggregate, even though the raw Walker2D reward was higher than NEAT’s. [PLACEHOLDER: final experiment per-task breakdown. Update Table 2 and this paragraph.]

The training curves show [PLACEHOLDER: describe from Figure 2 once generated, e.g. whether Walker2D convergence lags Hopper, whether HA-NEAT’s Hopper advantage appears early or late in training]. But these numbers are single-episode training estimates, and Section 5.4 shows they can be misleading.

5.4 Training fitness vs re-evaluation

Each genome is evaluated on a single episode per generation during training. That is a noisy estimate, especially for Walker2D, where episode outcomes depend on the initial physics state and contact sequence. To measure how noisy, we re-evaluated the best saved genomes from v4 over 5 episodes each, using different random seeds.

The gaps were large for HA-NEAT seed123 (Table 2, Figure 3). Its training-best Hopper fitness of 1370 dropped to 752 $\pm$ 435 under re-evaluation, a 45% reduction with massive episode variance. Walker2D held up somewhat better: 2330 at training, 1953 $\pm$ 483 at re-evaluation. NEAT seed42 was the opposite: re-eval scores were nearly identical to training (Hopper: 1200 vs 1196 $\pm$ 3; Walker2D: 1997 vs 1987 $\pm$ 11). This genome found a stable, low-variance gait. HA-NEAT seed42 fell in between, with a moderate Hopper drop (1905 to 1860 $\pm$ 322) and a larger Walker2D drop (2365 to 2005 $\pm$ 162).

The training-best fitness stored in MLflow is an optimistic single-episode snapshot. For HA-NEAT genomes, multi-episode re-evaluation can change the picture substantially. Section 6.4 discusses the implications.

5.5 Species dynamics and genome complexity

HA-NEAT changes speciation dynamics as a side effect of its design. When a node’s activation is mutated, all connections touching that node get new historical markers, which pushes the mutant into a new or isolated species. This could produce systematically more species than standard NEAT under otherwise identical settings. We compare species count and best-genome structural complexity over generations for both methods.

[PLACEHOLDER: Describe Figure 4 findings. Key question: does HA-NEAT produce more species than NEAT? Prior EDA on v2 data found a positive correlation between species count and fitness (higher species count $\rightarrow$ better outcomes). If HA-NEAT’s marker reassignment inflates species count, this could be a confound in attributing performance gains to activation diversity specifically.]

[PLACEHOLDER: Describe Figure 5 findings. Does HA-NEAT converge to simpler or more complex network topologies than NEAT? Activation diversity might reduce the pressure to add hidden nodes — one activation change can transform a node’s computational role without topological change.]

Whether these species and complexity patterns reflect a real advantage or just a confound is the question the ablation (Section 5.6) was designed to answer.

5.6 Ablation: activation diversity vs HA-NEAT machinery

HA-NEAT changes two things relative to standard NEAT: it draws per-node activations from $\{\tanh, \text{sigmoid}, \text{relu}, \sin, \text{identity}\}$, and it uses OriginConn with historical marker reassignment on activation mutation. The performance difference in Section 5.2 could come from either change or their interaction. We ran an ablation that keeps the full HA-NEAT infrastructure but restricts it to a single activation function. This is HA-NEAT machinery without the diversity.

Table 3. Ablation results: aggregate normalized-minimum fitness. Pop=1024, 500 generations, 3 seeds. HA-NEAT ablation uses HA-NEAT infrastructure with single activation.

Method	Seed 42	Seed 75	Seed 123	Mean
NEAT	0.414	0.342	0.381	0.379
HA-NEAT ablation (single activation)	0.352	0.342	0.398	0.364

The HA-NEAT machinery without activation diversity averaged 0.364 across three seeds, slightly below the NEAT baseline of 0.379 at the same configuration. The infrastructure alone did not help. If anything, it hurt slightly. Activation diversity appears to be what matters. [PLACEHOLDER: include full HA-NEAT

bar from final experiment data for direct comparison. Note the population size mismatch (ablation: 1024, main comparison: 2048+) when interpreting the magnitude of differences.]

5.7 Aggregation mode selection

How fitness is aggregated across tasks determines what the evolutionary pressure actually selects for. We tried three options in an initial sweep (36 runs, pop {512, 1024, 2048}, gen {100, 200}, seed 123): `normalized_sum` (weighted average), `normalized_product` (heavily penalizes near-zero scores on any task), and `normalized_min` (fitness equals the worst-performing task).

`normalized_product` produced the highest single-task score in the entire sweep: `neat_normalized_product_pop1024_gen200_seed123` reached 3606 on Walker2D (0.721 normalized), but Hopper was only 1096 (0.365 normalized). The genome had learned to exploit Walker2D while mostly ignoring Hopper. `normalized_sum` showed similar one-sided specialization. Under `normalized_min`, networks could not get away with this. The best `normalized_min` run (`ha_neat_normalized_min_pop2048_gen200_seed123`) scored Hopper=1182 and Walker2D=2012, both tasks progressing roughly in parallel. All subsequent experiments used `normalized_min`.

Chapter 6: Discussion

6.1 Main finding

Across the experiments reported above, HA-NEAT with heterogeneous activations produced higher aggregate multi-task fitness than NEAT with uniform tanh. The ablation points to activation diversity as the source rather than the OriginConn/marker infrastructure. But the evidence has gaps: 2-3 seeds per condition in the main comparison, substantial episode-level variance in HA-NEAT genomes on re-evaluation, and an ablation run at a smaller population size than the main experiment. HA-NEAT being a better multi-task optimizer than NEAT is plausible from this data. It is not proven.

6.2 Activation diversity and multi-task generalization

The preliminary data suggest heterogeneous activations help, though the mechanism is not directly observable from fitness curves.

A plausible explanation is that activation diversity expands the function space accessible without topological changes. A NEAT network with only tanh activations must add nodes or connections to change its computational behavior. An HA-NEAT network can shift a node from saturating (tanh) to rectified (relu)

without adding structure. In a multi-task setting where the same weights process 11-dimensional Hopper inputs and 17-dimensional Walker2D inputs, this allows the topology to be reused more efficiently across tasks.

Gaier and Ha (2019) showed in Weight Agnostic Neural Networks that varying the activation function globally across a fixed topology can solve diverse tasks, but they searched over activations at the architecture level, not per-node. Stanley and Miikkulainen’s original NEAT (2002) used a single fixed activation (sigmoid or tanh). Per-node diversity in NEAT variants has been tried, but rarely on multi-task physical simulation problems. [PLACEHOLDER: add citation for any multi-task NEAT variants in the literature if they exist; check ALIFE/GECCO proceedings.]

A separate issue is the Walker2D variance. NEAT seed42’s re-evaluation shows near-zero variance (std of 3-11 reward units), while HA-NEAT seed123 shows 435 on Hopper and 483 on Walker2D. HA-NEAT genomes may occupy a qualitatively different part of policy space, one that is more sensitive to initial conditions rather than strictly better. The training-best fitness comparison in Section 5.2 likely overstates HA-NEAT’s advantage because it relies on the single best episode seen during training.

6.3 The role of speciation in HA-NEAT

When HA-NEAT mutates a node’s activation, it reassigns historical markers on all connections touching that node. This pushes the mutant into its own species and shields it from crossover dilution. A side effect: HA-NEAT may produce more species than NEAT under identical settings. The v2 EDA found a positive correlation between species count and final fitness across NEAT runs, so if HA-NEAT’s marker mechanism systematically inflates species count, the performance advantage attributed to activation diversity could be partially confounded by a speciation effect.

The ablation speaks to this. HA-NEAT machinery without activation diversity (Section 5.6) did not outperform NEAT; it slightly underperformed. If the marker mechanism alone were the driver, producing more species that explore more of the search space, we would expect the ablation to at least match NEAT. It did not. The diversity itself appears necessary. [PLACEHOLDER: check Figure 4 (species dynamics) to verify whether HA-NEAT ablation also produces more species than NEAT. If yes, and it doesn’t help, this strengthens the activation-diversity interpretation.]

6.4 Training stochasticity and evaluation reliability

Section 5.4 exposed a failure mode worth discussing on its own: a genome that scores well on one episode may have gotten lucky rather than learned a robust policy. NEAT seed42’s near-zero re-evaluation variance is consistent with a stable, somewhat stereotyped gait. It always does roughly the same thing. HA-NEAT seed42’s moderate variance suggests a policy that is more sensitive to initial conditions, sometimes doing better and sometimes doing worse.

This matters for the performance claims. The MLflow `best_fitness` in Table 1 is the single best episode seen during training. For HA-NEAT seed42 in v4, the training-best Walker2D was 2365, but the 5-episode re-evaluation gave 2005 ± 162 , a 15% drop. For HA-NEAT seed123, Hopper went from 1370 to 752, a 45% drop. Any final comparison between methods needs multi-episode re-evaluation, not just the training checkpoint number.

6.5 Limitations

The main comparisons in Sections 5.2–5.3 rest on 2–3 seeds per condition. With this sample size it is not possible to compute meaningful confidence intervals or run significance tests. The final experiment uses 5 seeds, which is better but still at the low end for detecting moderate effect sizes — any performance difference below ~ 0.05 normalized fitness units should be treated as noise given the observed cross-seed variance.

Both NEAT and HA-NEAT used the same `compatibility_threshold`, `max_stagnation`, `survival_threshold`, and population size. HA-NEAT introduces `activation_mutate_rate` (fixed at 0.1 across all runs), which was not swept in the final experiment. It is possible that NEAT would benefit from different speciation settings than HA-NEAT, or that HA-NEAT’s activation mutation rate is suboptimal. The comparison is between default-configured methods, not optimally-configured ones.

Hopper and Walker2D are both bipedal locomotion tasks on flat terrain with the same reward function structure (forward velocity + control cost + alive bonus) and the same physics backend. They differ in dimensionality and difficulty, but they are not qualitatively different problem types. Generalization claims do not extend beyond this task family; it is unknown whether the findings hold for tasks with different objectives (manipulation, navigation) or substantially different physics.

The fitness estimate used during training is a single episode per genome. This affects both methods, but Section 5.4 shows it may affect HA-NEAT more if HA-NEAT genomes have higher episode variance. The training dynamics in Figures 1–2 reflect this noisy signal, not true expected policy performance.

Finally, the experiment went through several iterations (v1 through final) as early

results revealed problems: floor-penetration exploits in the generalized backend, aggregation mode imbalance, and under-powered population sizes and generation counts. The hypothesis was refined alongside the data. This is common in empirical ML work, but it means the results should be read as exploratory rather than confirmatory.

6.6 Future work

The most immediate next step is more environments. Ant adds 3D movement, HalfCheetah is a faster locomotion task with simpler contacts, and Humanoid is high-dimensional and unstable. Running the same comparison on 3 or 4 tasks would test whether the activation diversity advantage scales or was specific to the Hopper/Walker2D pair.

More seeds would also help. With 10-20 seeds per condition, Wilcoxon signed-rank tests and effect sizes with confidence intervals become feasible. Five seeds is enough to see trends but not enough to make strong claims.

HA-NEAT’s `activation_mutate_rate` was fixed at 0.1 without sweeping. A comparison of {0.05, 0.10, 0.20} would test whether 0.10 is actually a good value or just an arbitrary default. NEAT may also need a different `compatibility_threshold` than HA-NEAT to produce comparable species counts.

The `.npz` checkpoints contain the per-node activation assignments for every trained HA-NEAT genome, but we have not analyzed them. Do certain activations cluster near input nodes? Do genomes that score well on Walker2D favor relu over tanh in specific positions? This kind of analysis would add mechanistic evidence to what is currently just a performance comparison.

A comparison against gradient-based multi-task RL (multi-task SAC, MTRL variants) would put the neuroevolution numbers in a broader context. Gradient methods are not constrained to a single fixed-topology network, so the comparison is not apples-to-apples, but knowing the rough performance gap would be useful.

6.7 Closing

HA-NEAT extends NEAT with per-node activation diversity and a speciation mechanism that protects activation-mutant genomes. On a two-task Brax locomotion problem, the preliminary evidence suggests activation diversity provides a measurable benefit over uniform tanh, and the ablation puts the source of that benefit on the diversity rather than the OriginConn machinery. The final experiment will confirm or complicate this picture. But regardless of how that comparison resolves, the re-evaluation results are the more general finding: single-episode training fitness in NEAT-based RL is a noisy proxy, and performance

numbers reported from a single saved checkpoint should always be accompanied by multi-episode evaluation before any claim is made.